

1. Given the matrices:

$$A = \begin{pmatrix} -1 & 23 & 10 \\ 0 & -2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} -6 & 2 & 10 \\ 3 & -3 & 4 \\ -5 & -11 & 9 \\ 1 & -1 & 9 \end{pmatrix}, \quad C = (-3 \quad 2 \quad 9 \quad -5 \quad 7)$$
$$D = \begin{pmatrix} -2 & 6 \\ -5 & 2 \end{pmatrix}, \quad E = (3), \quad F = \begin{pmatrix} 3 \\ 5 \\ -11 \\ 7 \end{pmatrix}, \quad G = \begin{pmatrix} -6 & -4 & 23 \\ -4 & -3 & 4 \\ 23 & 4 & 9 \end{pmatrix}.$$

- (a) What is the dimension of each matrix?
- (b) Which matrices are square?
- (c) Which matrices are symmetric?
- (d) Which matrix has the entry at row 3 and column 2 equal to -11?
- (e) Which matrices has the entry at row 1 and column 3 equal to 10?
- (f) Which are column matrices?
- (g) Which are row matrices?
- (h) Find A^T, C^T, E^T and G^T .

2. Given the matrices:

$$A = \begin{pmatrix} 23 & 10 \\ 0 & -11 \end{pmatrix}, \quad B = \begin{pmatrix} -6 & 0 & 0 \\ -1 & -3 & 0 \\ -5 & 3 & -9 \end{pmatrix}, \quad C = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix},$$
$$D = \begin{pmatrix} -7 & 3 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & 9 \end{pmatrix}, \quad E = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 23 & 0 \\ 0 & 0 & -19 \end{pmatrix}.$$

- (a) Which of the above matrices are diagonal?
- (b) Which of the above matrices are lower triangular?
- (c) Which of the above matrices are upper triangular?

3. A, B, C, D and E are matrices with the orders

$$A : 2 \times 3, B : 3 \times 5, C : 5 \times 1, E : 1 \times 5$$

Which of the following are defined?

- (a) AB
- (b) AC
- (c) CE
- (d) EC
- (e) $(AB)C$

4. Given the matrices:

$$A = \begin{pmatrix} -1 & 1 & -2 \\ 0 & -2 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 2 & 0 \\ 0 & -3 & 4 \\ -1 & -2 & 3 \end{pmatrix}, \quad C = (-3 \quad 2 \quad 9 \quad -5 \quad 7),$$
$$D = \begin{pmatrix} -2 & 6 \\ -5 & 2 \end{pmatrix}, \quad E = \begin{pmatrix} 3 \\ 5 \\ -11 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & 2 \\ -2 & -3 & 4 \\ 1 & 4 & -3 \end{pmatrix}.$$

Find if possible:

- (a) AB
- (b) BC
- (c) AD
- (d) EF
- (e) FE

5. Find x and y if

$$A = \begin{pmatrix} x+y & -2 \\ x-y & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 12 & -8 \end{pmatrix}$$

6. Calculate

$$A = \left(\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix} \right)^{10}$$